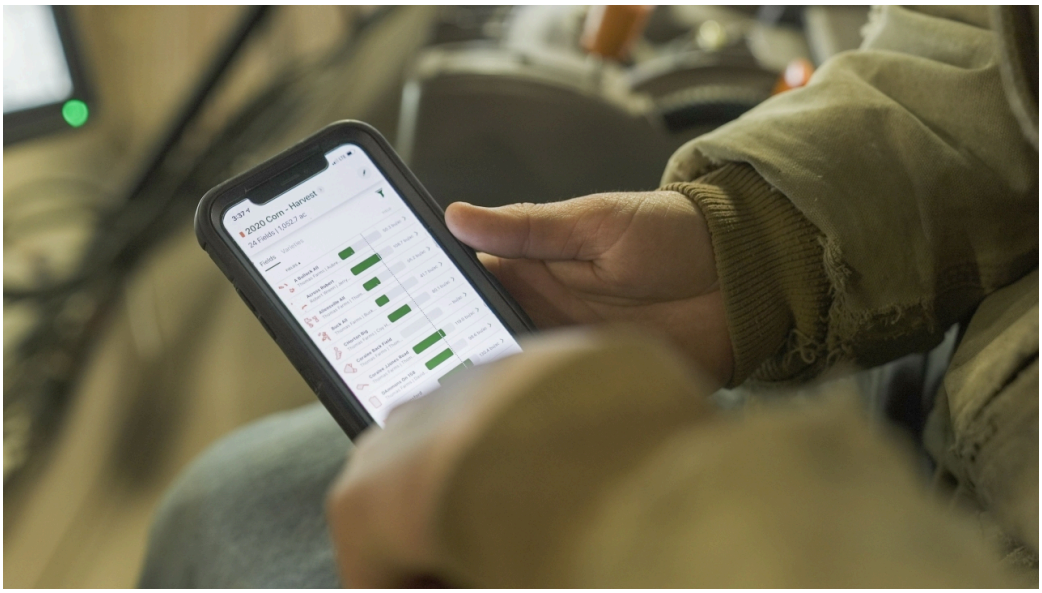


A farmer uses a smartphone to analyze crop data. Technological advancements have brought many benefits to agricultural and food systems, but they also pose unique risks that cybersecurity researchers say must be addressed. (Courtesy of Pexels)

A Nationwide Consortium Is Needed to Address Cybersecurity Risks in Agricultural and Food Systems, Researchers Urge

by College of Engineering Communications | October 08, 2026

Advances in technology have transformed America's agricultural and food systems. These innovations have brought benefits to production, flavor and disease resistance, but have also brought with them great vulnerabilities to cyberattacks.

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a paper in Nature Food outlining a path forward for cyber resiliency and robustness in agriculture. The paper recommends creating a national, geographically distributed Agriculture Cybersecurity Consortium, or ACC, to secure the nation's critical agricultural and food production complex.

The ACC would bring together farms, cooperatives, agribusinesses, cybersecurity businesses, government, nongovernmental organizations and academia to solve cybersecurity challenges as well as provide outreach and education programs. The bedrock of the ACC would be land-grant universities with strong computing and cybersecurity expertise, who would lead new initiatives in ag-centric cybersecurity research, testbeds, and education and workforce development.

The recommendation reflects a consensus perspective derived from three regionally coordinated workshops and follow-up sessions involving more than 150 stakeholders and participants. The effort was led by Iowa State University in collaboration with UC Davis, Washington State University, Virginia Tech University, North Carolina A&T University and the International Center for Food Ontology Operability Data and Semantics, or IC-FOODS, a non-profit research institute dedicated to designing Internet of Food cyberinfrastructure.

The Nature Food paper was overseen by Matthew Lange, CEO and Chief Science Officer of IC-FOODS, and Richard Hull, a principal scientist at IC-FOODS and computer science lecturer at UC Davis. Distinguished Professor Emeritus of Computer Science Matthew Bishop, Professor Emeritus of Computer Science Karl Levitt and Professor of Computer Science Mohammad Sadoghi participated in the conceptualization, review and editing process of the paper, contributing their expertise in cybersecurity.

"Every day, millions of small- and medium-sized agrifood operations across the supply chain face very real cyber risks, though many may not realize how vulnerable they are," Hull said. "With the advent of agentic AI, cyberattacks are growing in number and sophistication. Our hope with this paper is to accelerate a shift toward making cyberhealth a

The workshops discussed in the Nature Food paper were supported in part by funding from Iowa State's Presidential Interdisciplinary Research Initiative. Support for IC-FOODS came from National Science Foundation funding for the AI Institute for Intelligent Cyberinfrastructure with Computational Learning in the Environment, or ICICLE.

"Cybersecurity is not simply an IT problem — it is a food-system resilience problem," said Lange, CEO of IC-FOODS and an alum of UC Davis. "The Agriculture Cybersecurity Consortium gives us a way to build the collective capacity needed to confront that systemic risk."

Read Iowa State University's press release

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Matthew A. Bishop
Computer Science

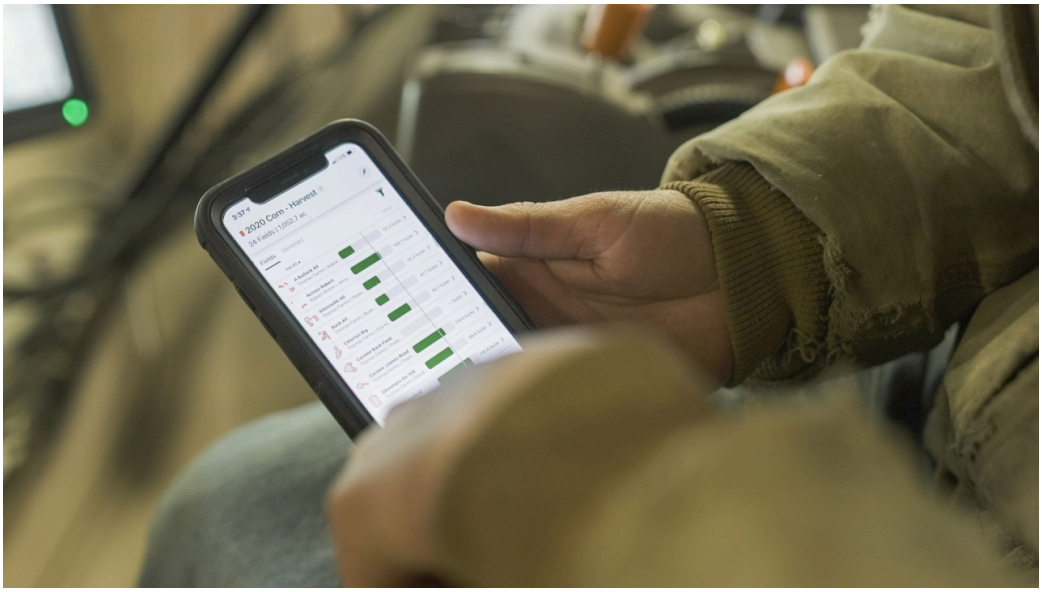


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